

Replicating Wide Residual Networks for Image Classification

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Introduction

Deep residual networks (ResNets) have traditionally emphasized scaling performance by increasing network depth. However, this extreme depth introduces the "diminishing feature reuse" problem, where many residual blocks contribute very little to the final goal. While identity mapping is a core strength of ResNets, it is also a weakness because there is no guarantee that gradients will pass through the residual blocks effectively.

This project re-implements the core findings of the 2016 paper, "Wide Residual Networks", by Sergey Zagoruyko and Nikos Komodakis. The authors propose Wide Residual Networks (WRNs) as an alternative approach, demonstrating that decreasing layer count while increasing the width of the network is a highly effective and computationally efficient way to improve model speed and accuracy.

Chosen Result

The specific result we aimed to reproduce was achieving an approximate 4% test error rate on the CIFAR-10 dataset using the WRN-28-10 architecture. More importantly, we sought to demonstrate that this shallower, wider network yields a ~0.9% decrease in error compared to the much deeper, 1000+ layer ResNet-1001.

This result was chosen because it represents the crux of the paper's thesis: it proves that WRNs effectively solve the diminishing feature reuse problem. By validating this metric, we can confirm that a well-designed wide network can dramatically outpace ultra-deep networks in both training time and final classification accuracy.

pre-act-ResNet[13]	110	1.7M	6.37	-
	164	1.7M	5.46	24.33
	1001	10.2M	4.92(4.64)	22.71
WRN (ours)	40-4	8.9M	4.53	21.18
	16-8	11.0M	4.27	20.43
	28-10	36.5M	4.00	19.25

Methodology

Model Architecture: The network begins with an initial 3x3 convolution producing 16 channels, followed by three main groups of residual blocks. The total number of blocks per group is defined by $N = (n-4)/6$. For our primary model, WRN-28-10, the depth is $n=28$ and the widening factor is $k=10$, resulting in 4 blocks per group (12 residual blocks total). The channel dimensions scale significantly across the groups based on the widening factor k , progressing from 16k channels in Group 1, to 32k in Group 2, and finally 64k in Group 3.

Dataset: We utilized the CIFAR-10 dataset, which consists of 60,000 images distributed across 10 classes, to train and test the models.

Training Protocol: We optimized the models using Stochastic Gradient Descent (SGD) with Nesterov momentum and cross-entropy loss. The initial learning rate was set to 0.1, with a weight decay of 0.0005 and momentum at 0.9. The learning rate was dropped by 0.2 at epochs 60, 120, and 160, running for a total of 200 epochs. To mitigate overfitting, we utilized batch normalization.

Modifications & Scope: Due to computational limits, we trained each network configuration once, rather than averaging over five runs as done in the original paper. Additionally, we scoped our project to exclude the dropout-in-residual-blocks experiments, the depth ablations, and the CIFAR-100, SVHN, and ImageNet benchmarks evaluated by the original authors.

Results & Analysis

Our re-implementation successfully reproduced the performance benchmarks established in the original paper, confirming the superiority of the widened architecture over purely deep models. When holding depth constant at 40 layers, increasing the widening factor k resulted in a consistent decrease in test loss. Our WRN-40-1 baseline achieved a 6.61% test error, which improved to 5.12% for WRN-40-2, and further dropped to 4.32% for WRN-40-4. In all of these trials, our models slightly outperformed the paper's reported error rates (e.g., WRN-40-4 achieved a 3.65% positive delta over the paper's 4.97% error). For our primary target, the WRN-28-10 model (36.48M parameters), we achieved a final test error of 3.93%. This closely matches the paper's reported 4.00% error. Crucially, this result firmly beats the baseline pre-activation ResNet-1001 (10.2M parameters), which reported a 4.92% test error. This validates our core hypothesis: the higher parameter, wider model achieves greater accuracy with better efficiency than thin, 1000-layer networks.

Reflections

This project provided significant insight into the practical tradeoffs between depth, width, and computational efficiency in deep learning. We observed firsthand that blindly scaling depth yields diminishing returns. Conversely, wider architectures align remarkably well with hardware optimization, enabling far more efficient parallel computation on GPUs.

A key challenge was managing computational resources. Training WRN-28-10 takes substantial time, which requires us to limit our experiments to single-run evaluations rather than computing medians over multiple runs.

For future work, we would implement dropout layers directly between the convolutions in the residual blocks. The original authors highly recommended this technique to further prevent overfitting in wider networks. Additionally, testing the models on the more complex CIFAR-100 dataset would allow us to better analyze the upper limits of the widening factor.

References

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- Krizhevsky, A., Nair, V., & Hinton, G. (2012). CIFAR-10 (Canadian Institute for Advanced Research). <http://www.cs.toronto.edu/~kriz/cifar.html>